W-T-001567(GSA-FSS)
July 1, 1975

INTERIM FEDERAL SPECIFICATION

TAPE, RECORDING, QUADRUPLEX VIDEO, MAGNETIC OXIDE-COATED

This Interim Federal Specification was developed by the General Services Administration, Federal Supply Service, Washington, DC 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the detailed requirements for 2-inch laterally oriented magnetic recording tape wound on reels, designed for use on quadruplex, transverse-scan, broadcast type video recorder-reproducers.

1.2 Classification. Tape covered by this specification is designated in the following form, as specified (see 6.1):

TVT	B	HT	2000	PCP	II	1200
Basic indicator (see 1.2.1)	Oxide characteristic indicator (see 1.2.2)	Backing and thickness indicator (see 1.2.3)	Width indicator (see 1.2.4)	Reel and hub indicator (see 1.2.5)	Reel type indicator (see 1.2.6)	Length indicator (see 1.2.7)

1.2.1 Basic indicator. The basic indicator defines the general application for a particular class of tape as specified herein. The basic indicator is "TVT" for tape, video type.

1.2.2 Oxide characteristic indicator. The oxide characteristic indicator identifies a particular oxide by its physical or magnetic characteristics as specified herein. The oxide characteristic indicator "B" denotes tape intended for use on quadruplex, transverse-scan video tape recorders which have the capability of recording video, audio and tracking control signals in accordance with ANSI Standards and SMPTE Recommended Practices (see 2.2).

1.2.3 Backing and thickness indicator. The backing and thickness indicator combines two letters as an indicator. The first letter identifies the type of backing material and the second letter indicates the nominal thickness of the backing. The backing material is indicated by the letter "H" for polyester base film. The backing material of a nominal thickness of 0.8 to 1.0 mils (thousandths of an inch) is defined by the letter "T" for "thin".

1.2.4 Width indicator. The width indicator is the nominal width of the tape in thousandths of an inch (2 inch tape is 2000), (see 3.2.2).

1.2.5 Reel and hub indicator. The reel and hub indicator is identified by letters in accordance with 3.1.3.4.

1.2.6 Reel type indicator. The reel type indicator is identified by letters in accordance with 3.1.3.4.

1.2.7 Length indicator. The length indicator is the minimum length, in feet, of a tape wound on a reel (see 3.2.1).

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2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on the date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

L-P-378 - Plastic Sheet and Strip Thin Gauge (Polyolefin).
W-R-175/GEN - Reels and Hubs for Magnetic Recording Tape, General Specification for.
W-R-175/4 - Reels, Precision, Aluminum, 3-inch Center Hole.
PPP-B-636 - Boxes, Shipping, Fiberboard.

Federal Standards:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).
Fed. Test Method Std. No. 406/GEN - Plastics: Methods of Testing.
Fed. Test Method Std. No. 406/1013 - Tensile Properties of Thin Sheets and Films.
Fed. Test Method Std. No. 406/4041 - Electrical Resistance (Surface).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.)

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from the established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids, or request for proposal, shall apply:

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Tariff Order Section, 1616 P Street, NW, Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

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American Society for Testing and Materials (ASTM) Standards:

D 638 - Tensile Properties of Plastics: Method of Test for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

American National Standards Institute (ANSI) Documents:

C98.1 - Dimensions of 2-Inch Video Magnetic Tape
C98.3 - Electrical Characteristics of Audio Record One for 2-Inch Quadruplex Video Magnetic Tape Recording at 15 and 7.5 in./s.
C98.4 - Speed of 2-Inch Tape for Quadruplex Video Magnetic Tape Recording.
C98.6 - Video, Audio and Tracking Control Records on 2-Inch Video Magnetic Tape.
C98.7 - Specifications for a Primary Audio Reference Level Recording for Quadruplex Video Magnetic Tape Recorders Operating at 15 in./s.
C98.8 - Specifications for an Audio Level and Multifrequency Test Tape for Quadruplex Video Magnetic Tape Recorders Operating at 15 in./s/
C98.10- Specifications for a Primary Audio Reference Level Recording for Quadruplex Video Magnetic Tape Recorders Operating at 7.5 in./s.
C98.11- Specifications for an Audio Level and Multifrequency Test Tape for Quadruplex Video Magnetic Tape Recorders Operating at 7.5 in./s.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

Society of Motion Picture and Television Engineers (SMPTE) Recommended Practices (RP):

RP 6 - Reference Carrier Frequencies and De-Emphasis Characteristics for 2-Inch Quadruplex Video Magnetic Tape Recording.
RP 10 - Video Alignment Signal Specifications for Quadruplex Video Magnetic Tape Recording.
RP 11 - Tape Vacuum Guide Radius and Position for 2-Inch Quadruplex Video Magnetic Tape Recording.
RP 16 - Specifications of Tracking Control Record for 2-Inch Quadruplex Video Magnetic Tape Recording.
RP 29 - Video Test Tape for Quadruplex Video Frequency Magnetic Tape Recorders Operating at 15 in./s. and Practice LBM of SMPTE Recommended Practice RP6.
RP 30 - Video Test Tape for Quadruplex Video Frequency Magnetic Tape Recorders Operating at 7.5 in./s. and Practice LBM of SMPTE Recommended Practice RP6.
RP 31 - Video Test Tape for Quadruplex Video Frequency Magnetic Tape Recorders Operating at 15 in./s. and Practice LBC of SMPTE Recommended Practice RP6.
RP 43 - Video Test Tape for Quadruplex Video Frequency Magnetic Tape Recorders Operating at 15 in./s. and Practice HB of SMPTE Recommended Practice RP6.
RP 44 - Video Test Tape for Quadruplex Video Frequency Magnetic Tape Recorders Operating at 7.5 in./s. and Practice HB of SMPTE Recommended Practice RP6.
RP 47 - Electronic Method of Dropout Detection and Counting.

(Application for copies should be addressed to the Society of Motion Picture and Television Engineers, 862 Scarsdale Ave., Scarsdale, NY 10583).

3. REQUIREMENTS

3.1 General requirements.

3.1.1 Product qualification. All video magnetic tape supplied under this specification shall be products which have been tested and approved for listing on the applicable Qualified Products List (QPL) maintained by the General Services Administration (GSA-FSS). Hereinafter the term qualifying activity shall mean the GSA.

3.1.1.1 Qualified products listing. The Qualified Products List shall consist of products which have been tested and passed those qualification tests specified herein.

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3.1.1.2 Qualification product testing. Products may be submitted by prospective suppliers of magnetic tape on dates to be announced by the General Services Administration. Should the submitted product(s) fail to meet the specification as defined herein, the qualifying activity reserves the right to refuse to accept such products for additional qualification tests until satisfactory data and test results have been submitted indicating correction of product deficiencies. The qualifying activity reserves the right to levy a charge to cover cost of testing for qualification of products.

3.1.1.3 Qualification withdrawal. A supplier's qualified product listing may be withdrawn in accordance with "Provisions Governing Application by Manufacturers for Inclusion on Federal Qualified Products Lists (QPL's)" (see 6.3).

3.1.1.4 Re-qualification. A supplier's product, once withdrawn from the Qualified Products List, shall not be accepted for re-qualification until satisfactory data and test results have been submitted to the qualifying activity by the supplier indicating correction of the product fault(s). The qualifying activity reserves the right to levy a charge to cover cost of testing for re-qualification of products.

3.1.1.5 Process change. Qualification of a supplier's product under this specification, once established, applies only to those tapes manufactured by the specific process in use at the time of qualification. All proposed process changes shall be reported to the qualifying activity with a statement by the supplier as to the general nature of the changes and the extent and effect of such changes on the delivered product. The qualifying activity reserves the right to require six weeks notice, and sufficient samples of the new product for qualification tests, from the supplier prior to delivery of products manufactured under any such change in process. The foregoing requirement does not in any way relieve the manufacturer of delivery requirements, quality control and testing necessary to insure that all products delivered under this specification are equal to those products originally submitted for qualification.

3.1.2 Component parts. Tape delivered under this specification shall be supplied on reels that meet the requirements of 3.1.3.4 and that are packaged in accordance with 5.1, as specified (see 6.1).

3.1.3 Materials.

3.1.3.1 Tape. The tape supplied under this specification shall be new and shall consist of a layer of magnetic material on a suitable backing material and shall have the physical, magnetic and other characteristics as specified herein.

3.1.3.2 Toxic compounds. The tape shall not cause bodily harm by contact, inhalation or ingestion under conditions normally encountered in Government Service.

3.1.3.3 Flammable materials. Tape which will ignite from a match flame, and when so ignited continue to burn in a still carbon dioxide atmosphere, shall not be supplied under this specification.

3.1.3.4 Reels. Reels shall conform to the requirements of W-R-175, unless otherwise specified (see 6.1).

3.1.4 Wind. The tape shall be wound on the reel with the oxide surface facing toward the hub.

3.2 Dimensional requirements.

3.2.1 Length. The tape shall be supplied in minimum lengths in accordance with table I, unless otherwise specified (see 6.1), on reels that conform to the requirements of 3.1.3.4.

3.2.2 Width. The width of the tape shall be $2.000 \pm \begin{smallmatrix} 0.000 \\ 0.004 \end{smallmatrix}$ inches.

3.2.3 "E" value. The minimum "E" value for all tape lengths shall be not less than 0.2 inches. The "E" value is defined as the radial distance by which the reel flanges extend beyond the outermost layer of tape when the tape is wound at a tension of 14 ounces $\pm$ 4 ounces on a reel that meets the requirements of 3.1.3.4.

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TABLE I. Tape Length

Nominal reel diameter (inches)	Length indicator
8.0	1200 1650
10.5	1200 1650 2550 3600
12.5	2550 3600 4950 5550
14.0	7200

3.3 Physical requirements.

3.3.1 Yield strength. The 1 percent offset yield point, the force at 3 percent elongation and the breaking force of the tape shall equal or exceed the value specified in table II when tested in accordance with 4.4.4.

3.3.2 Elongation under stress. The elongation of the tape shall not exceed the value specified in table II when tested in accordance with 4.4.5.

3.3.3 Layer-to-layer adhesion. The tape shall show no sticking or layer-to-layer adhesion when tested in accordance with 4.4.6. (Layer-to-layer adhesion, commonly termed blocking, is defined as that property of a magnetic tape wherein one layer when held in close proximity to an adjacent layer, as when in a reel of tape, exhibits an adhesive nature and bonds itself to an adjacent layer so that free and smooth separation of the layers is difficult. In practice, the amount of adhesion may vary from a quantity which merely prevents adjacent layers from separating freely to the most severe condition in which the adhesion is so great that the magnetic coating is disengaged from the tape backing when the tape is unwound. Adhesion in any form is objectionable, since tape which displays even mild layer-to-layer adhesion may in time progress to the point where it is impossible to separate layers without destroying the tape. In operation, adhesion can result in excessive flutter, timing errors and poor uniformity of output level.)

3.3.4 Longitudinal curvature. The deviation of each tape edge from a straight line shall not exceed the value specified in table II when measured in accordance with 4.4.7.

3.3.5 Wound pile envelope. The tape as initially received from the manufacturer shall be smoothly wound to form an integral mass and shall not be loose either axially or circumferentially. There shall be no visible folds, buckling, cinching, spoking, or gaps between the tape layers. The edges of the wound tape shall be in a single plane with a minimum of protruding tape edges, layers, or groups of layers. The tape shall be wound in such a manner that it will not be physically damaged on either edge resulting in functional failure (see 4.4.8). In addition, the tape shall not exhibit any of the above defects, which result in functional failure, during all normal operating modes (fast forward, rewind, etc.) on the reference test recorder.

3.3.6 Anchorage. The magnetic layer of the tape shall show no visible evidence of anchorage failure after completion of all tests (see 4.4.9). Anchorage failure may be manifested as a separation of the magnetic coating from the base material of the tape, or internal failure of the coating layer itself. Commonly used recorder solvents/cleaners shall have no effect on the tape during subsequent passes on the reference test recorder.

3.3.7 Splices. All tape lengths supplied under this specification shall be continuous and splice free (see 4.4.10).

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3.3.8 Workmanship and general examination. The magnetic tape, reels, and containers shall be manufactured and processed in a careful and workmanlike manner in accordance with good practice. Both the magnetic oxide surface and base shall be free from any raised areas, dust, flakes, powder, holes, foreign matter, scratches, creases and other defects which would render the tape unsuitable for its intended use. The condition of tape slitting, reels, wind, and containers shall be examined for workmanship and cleanliness (see 4.4.11).

TABLE II. Physical requirements

Characteristic	Backing and thickness indicator (see 1.2.3) HT	Units (per 2-inches of tape width)
Yield strength	26.0	pounds
Elongation under stress	0.50	percent
Longitudinal curvature	0.0625	inches

3.4 Magnetic requirements.

3.4.1 Sensitivity.

3.4.1.1 RF output sensitivity. The RF output level of the tape at any point in a reel, when measured in accordance with 4.5.3.1.1, shall not vary from the standard RF output level by more than the value specified in table III.

3.4.1.2 Audio sensitivity. The audio output level of the tape at any point in a reel, when measured in accordance with 4.5.3.1.2, shall not vary from the standard audio output level by more than the value specified in table III.

3.4.2 Frequency response.

3.4.2.1 Audio frequency response. The output levels of the tape at the specified test frequencies, when measured in accordance with 4.5.3.2.1, and when normalized to the 1,000 Hz output level and compared with the response of the reference tape, shall fall within the limits specified in table III.

3.4.3 Instantaneous nonuniformity (dropouts). The average number of dropouts per minute and the maximum number of dropouts in any one minute, when tested as specified in 4.5.3.3, shall not exceed the values specified in table III.

3.4.4 Output level uniformity - short term.

3.4.4.1 Audio uniformity. The uniformity of the signal output at test frequencies of 1,000 Hz and 8,000 Hz, when tested as specified in 4.5.3.4.1, shall be such that the maximum deviation from the maximum peak output in any 10 second interval does not exceed the values specified in table III.

3.4.4.2 Control track uniformity. The uniformity of the signal output at the specific test frequency, when tested as specified in 4.5.3.4.2, shall be such that the maximum deviation from the maximum peak output in any 10 second interval does not exceed the value specified in table III.

3.4.4.3 Cue track uniformity. The uniformity of the signal output at the specific test frequency, when tested as specified in 4.5.3.4.3, shall be such that the maximum deviation from the maximum peak output in any 10 second interval does not exceed the value specified in table III.

3.4.5 Output level uniformity - long term.

3.4.5.1 Audio uniformity. The uniformity of the signal output at test frequencies of 1,000 Hz and 8,000 Hz, when tested as specified in 4.5.3.5.1, shall be such that the variation from maximum peak output to minimum peak output over the entire recorded tape length does not exceed the values specified in table III.

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3.4.5.2 Control track uniformity. The uniformity of the signal output at the specific test frequency, when tested as specified in 4.5.3.5.2, shall be such that the variation from maximum peak output to minimum peak output over the entire recorded tape length does not exceed the value specified in table III.

3.4.5.3 Cue track uniformity. The uniformity of the signal output at a test frequency of 1,000 Hz, when tested as specified in 4.5.3.5.3, shall be such that the variation from maximum peak output to minimum peak output over the entire recorded tape length does not exceed the value specified in table III.

3.4.6 Harmonic distortion. The third harmonic distortion at the specific test record levels, when measured in accordance with 4.5.3.6.1 and 4.5.3.6.2, shall not exceed the values specified in table III.

3.4.7 Signal-to-noise ratio.

3.4.7.1 Video signal-to-noise ratio. The video signal-to-noise ratio of the tape, when tested in accordance with 4.5.3.7.1, shall not be less than the video signal-to-noise ratio of the reference tape by more than the value specified in table III.

3.4.7.2 Audio signal-to-noise ratio. The audio signal-to-noise ratio of the tape, when tested in accordance with 4.5.3.7.2, shall not be less than the audio signal-to-noise ratio of the reference tape by more than the value specified in table III.

3.4.8 Layer-to-layer signal transfer. A signal resulting from layer-to-layer signal transfer, when measured as specified in 4.5.3.8.1 shall be below the signal level by at least the value specified in table III.

3.4.9 Ease of erasure.

3.4.9.1 Audio erasure. The specified erase field shall effect a reduction in level of a 1000 Hz signal of at least the value specified in table III, when tested as specified in 4.5.3.9.1.

3.4.10 Electrical resistance. The electrical resistance of either surface of the tape (the oxide surface or the back side surface), when tested as specified in 4.5.3.10, shall not exceed the value specified in table III.

3.4.11 Durability. When subjected to the wear tests specified in 4.5.3.11, the audio uniformity-short term and long term, the control track uniformity-short term, and the dropout characteristic of the tape shall meet the requirements of 3.4.4.1, 3.4.5.1, 3.4.4.2, 3.4.5.2, and 3.4.3, respectively. In addition, the variation in RF output level shall not exceed ± 3 db when tested in accordance with 4.5.3.11.

3.4.12 Abrasivity. The abrasivity characteristic of the tape shall be such that when tested as specified in 4.5.3.12, the average head wear rate shall not exceed the value specified in table III.

3.4.13 Environmental extremes. When subjected to the temperature and humidity conditions specified in 4.5.3.13, the tape shall meet the requirements of Instantaneous nonuniformity (see 3.4.3), Output level uniformity - short term (see 3.4.4.1 and 3.4.4.2) and Output level uniformity - long term (see 3.4.5.1 and 3.4.5.2).

3.4.14 Frictional vibration. The tape shall not produce squeal components of an audible nature during the magnetic tests. Frictional vibration, which in practice manifests itself as a squeal, is defined as the random frequency and amplitude modulation, or both, of a recorded signal which results when the tape passes over the guides and heads of a magnetic tape recorder. This results from the changing frictional components introduced between the tape, heads and guides as the tape travels past them.

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TABLE III. Magnetic Requirements

Characteristic	Requirement	Units
Sensitivity		
RF output	0 ± 2	Decibels
Audio	0 ± 2	Decibels
Frequency response		
Audio		
200 Hz	0 ± 2	Decibels
5,000 Hz	0 ± 2	Decibels
8,000 Hz	0 ± 2	Decibels
15,000 Hz	0 ± 2	Decibels
Instantaneous nonuniformity (dropouts)	15	Dropouts/minute (avg.)
	30	Dropouts/minute (max.)
Output level uniformity-short term		
Audio		
1,000 Hz	2	Decibels
8,000 Hz	2	Decibels
Control track	2	Decibels
Cue track	2	Decibels
Output level uniformity-long term		
Audio		
1,000 Hz	2	Decibels
8,000 Hz	2	Decibels
Control track	2	Decibels
Cue track	3	Decibels
Harmonic distortion		
Std. audio record level	1.0	Percent
Max. audio record level	4.0	Percent
Signal-to-noise ratio		
Video	3 (see 3.4.7.1)	Decibels
Audio	2 (see 3.4.7.2)	Decibels
Audio layer-to-layer signal transfer	40	Decibels
Audio ease of erasure	60	Decibels
Electrical resistance	100	Megohms/square
Durability	see 3.4.11	----------
Abrasivity	10	Microinches/hour
Environmental extremes	see 3.4.13	----------

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

(a) Qualification testing (see 4.2.1).
(b) Quality conformance inspection (see 4.2.2).
(c) Acceptance inspection (see 4.2.3).
(d) Examination of preparation for delivery (see 4.2.4).

4.2.1 Qualification testing. Qualification testing shall be conducted at a laboratory designated by the General Services Administration, Federal Supply Service, and shall consist of tests listed in 4.3, 4.4 and 4.5. All samples subjected to these tests must conform to the associated requirements listed in section 3 of this specification, unless otherwise specified.

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4.2.1.1 Qualification of a particular tape base, thickness, and oxide characteristic shall constitute qualification of all tapes of that particular base, thickness and oxide characteristic regardless of length.

4.2.1.2 Samples for qualification. The sample size shall consist of ten (10) reels of 2-inch 2550-foot tape lengths for each particular tape base, thickness and oxide characteristic for which qualification is desired. All tape samples shall be submitted on reels which conform to the requirements of 3.1.3.4. The manufacturer shall also submit test data showing that each tape type for which qualification is desired has successfully met all the requirements of this specification.

4.2.2 Quality conformance inspection. In accordance with 4.1, the contractor shall be responsible for the performance of the quality conformance inspections. The contractor shall select a sample number of reels of tape from each lot of tape offered for delivery. Sample reels of tape shall be selected in accordance with 4.2.2.2 and the contractor shall subject the samples to the examinations and tests of table IV to determine compliance with the specification requirements. All necessary test equipment and facilities required to conduct the subject examinations and tests shall be furnished by the contractor. Alternate test techniques and equipment may be used by the contractor, subject to review and approval by the qualifying activity that such techniques and equipment are in fact adequate to insure compliance with specification requirements.

4.2.2.1 Reporting of quality conformance test results. Unless otherwise directed by the contracting officer (see 6.1), the contractor shall maintain a complete record of all production test results for the duration of the contract. The production test records shall be available to the Government at all reasonable times. The records shall include the information necessary to identify the lot, the reels of tape, the testing equipment, the inspector, and the date of test.

4.2.2.2 Sampling for quality conformance inspection. Sampling procedures for quality conformance inspection shall be in accordance with MIL-STD-105 and as specified herein. The sample reels of tape shall be subjected to the Group A, B, C and D examinations and tests shown in table IV.

4.2.2.2.1 Sampling for group A examinations and tests. The sampling procedure for group A examinations and tests shall be in accordance with MIL-STD-105, special inspection level S-2 at an acceptance quality level (AQL) of 10 percent defective.

4.2.2.2.2 Sampling for group B examinations and tests. The sampling procedure for group B examinations and tests shall be in accordance with MIL-STD-105, special inspection level S-1 at an acceptance quality level of 10 percent defective.

4.2.2.2.3 Sampling for group C examinations and tests. The sampling procedure for group C examinations and tests shall be in accordance with MIL-STD-105, special inspection level S-1 at an acceptance quality level of 10 percent defective.

4.2.2.2.4 Sampling for group D examinations and tests. Group D examinations and tests will be required initially and as often as deemed necessary to assure continued compliance with the specified requirements. Five sample reels from a lot shall be subjected to the group D examinations and tests shown in table IV. Failure to conform to any requirement of the group D examinations and tests shall be cause for the rejection of that lot represented by the samples.

4.2.3 Acceptance inspection. Acceptance testing on shipment lots shall be conducted by the qualifying activity. Acceptance tests shall consist of those group A and group B examinations and tests specified in table IV.

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4.2.3.1 Sampling for acceptance inspection. The sampling procedures for the group A and group B examinations and tests shall be as specified in 4.2.2.2.1 and 4.2.2.2.2.

TABLE IV. Inspection Tests

Test Group	Examinations and Tests	Requirement Paragraph	Test Paragraph
	Toxic Compounds	3.1.3.2	----------
	Wind	3.1.4	----------
	Length	3.2.1	4.3.1
	Width	3.2.2	4.3.2
	E value	3.2.3	4.3.3
A	Wound pile envelope	3.3.5	4.4.8
	Anchorage	3.3.6	4.4.9
	Splices	3.3.7	4.4.10
	Workmanship and general examination	3.3.8	4.4.11
	Instantaneous nonuniformity (dropouts)	3.4.3	4.5.3.3
	Output level uniformity-short term	3.4.4	4.5.3.4
	Audio uniformity	3.4.4.1	4.5.3.4.1
	Control track uniformity	3.4.4.2	4.5.3.4.2
	Output level uniformity - long term	3.4.5	4.5.3.5
	Audio uniformity	3.4.5.1	4.5.3.5.1
	Control track uniformity	3.4.5.2	4.5.3.5.2
	Sensitivity	3.4.1	4.5.3.1
B	Signal-to-noise ratio	3.4.7	4.5.3.7
	Frictional vibration	3.4.14	----------
	Yield strength	3.3.1	4.4.4
	Layer-to-layer adhesion	3.3.3	4.4.6
C	Frequency response	3.4.2	4.5.3.2
	Flammable materials	3.1.3.3	----------
	Reels	3.1.3.4	----------
	Elongation under stress	3.3.2	4.4.5
D	Longitudinal curvature	3.3.4	4.4.7
	Output level uniformity-short term	3.4.4	4.5.3.4
	Cue track uniformity	3.4.4.3	4.5.3.4.3
	Output level uniformity-long term	3.4.5	4.5.3.5
	Cue track uniformity	3.4.5.3	4.5.3.5.3
	Harmonic distortion	3.4.6	4.5.3.6
	Layer-to-layer signal transfer	3.4.8	4.5.3.8
	Ease of erasure	3.4.9	4.5.3.9
	Electrical resistance	3.4.10	4.5.3.10
	Durability	3.4.11	4.5.3.11
	Abrasivity	3.4.12	4.5.3.12
	Environmental extremes	3.4.13	4.5.3.13

4.2.4 Examination of preparation for delivery. Interior and exterior examinations shall be made to determine that the packaging, packing, and marking comply with the requirements in section 5, as specified (see 6.1). A sample unit for these examinations shall be one shipping container, fully prepared for delivery. The sample units shall be selected from end item inspection lots and examined for the defects listed in table V. The sampling procedure shall be in accordance with MIL-STD-105, Special Inspection Level S-2 at an acceptance quality level of 6.5 percent defective. The lot size shall be taken as the number of shipping containers in the end item inspection lot.

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TABLE V. Preparation for delivery defects

Examine	Defects
Interior packaging	Use of improper or defective materials.
	Quantity in unit container not as specified.
	Quantity in shipping container not as specified.
	Incorrect packaging method employed.
	Cushioning or padding omitted.
	Cushioning inadequate for the physical protection of the item.
	Unsealed, punctured, or improperly sealed bag, wrap, or envelope.
	Damaged or otherwise defective reels.
Exterior packing and marking	Use of improper or defective materials.
	Type, grade, class and style of shipping container not as specified.
	Marking incorrect; incomplete; illegible; omitted; of improper size, location, sequence, or method of application.
	Inadequate application of components such as: Incomplete closure of case liners, container flaps, loose or inadequate sealing, strapping or stapling.
	Bulged or distorted container.

4.3 Dimensional measurements.

4.3.1 Length. The tape length shall be measured to determine conformance to the requirements of 3.2.1.

4.3.2 Width. The tape width shall be measured to determine conformance to the requirements of 3.2.2.

4.3.3 "E" value. The "E" value shall be measured to determine conformance to the requirements of 3.2.3.

4.4. Physical tests and measurements.

4.4.1 Preliminary conditioning. Preliminary conditioning of each reel of tape will be required prior to qualification testing, when specified in the individual test paragraphs, to relieve stresses and establish uniformity. Reels of tape shall be rewound under a tension of 14 ounces ± 4 ounces. The reels of tape shall then be conditioned at 125° ± 5° F., 80 to 90 percent relative humidity (r.h.) for 3 hours. The tape shall then be conditioned at 75° ± 5° F., 50 ± 5 percent r.h. for a minimum of 3 hours. This temperature humidity cycle shall be repeated, with the transition times between temperature extremes not to exceed 60 minutes, until the tape has been subjected to a total conditioning time, including transitions, of 20 hours. Upon removal after the last cycle, the tape shall immediately be rewound a minimum of three times at a tension of 14 ounces ± 4 ounces. The tape shall then be held in a room conditioned at 75° ± 5° F., 50 ± 5 percent r.h. for a minimum of 24 hours before any tests are made. Preliminary conditioning shall not be required for quality conformance testing.

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4.4.2 Test conditions. Unless otherwise specified in the contract or order, all tests required by this specification shall be made at an ambient temperature of 75° ± 5° F., and a relative humidity between 45 and 55 percent in a relatively dust free atmosphere.

4.4.3 Test samples. All physical tests required by this specification shall be performed on 2 inch wide tape samples.

4.4.4 Yield strength.

4.4.4.1 Apparatus. The apparatus for determining the specified tensile properties shall be as described in paragraph 3.2 of Fed. Test Method Std. No. 406, Method 1013.

4.4.4.2 Preparation of samples. Samples shall be conditioned as specified in 4.4.1.

4.4.4.3 Procedure. A strip of tape not less than 7 inches in length shall be clamped in the grips set for an initial separation of 4 inches. The test specimen shall be clamped in the testing machine taking care to align the long axis of the specimen with an imaginary line joining the points of attachment of the grips to the machine. The grips shall be tightened evenly and firmly to the degree necessary to prevent slipping of the specimen during the test. The rate of grip separation shall be 2 inches per minute. Not less than five specimens of tape shall be tested. A force strain diagram shall be obtained. Means shall be provided for calibration of the strain axis. The 1.0 percent offset yield strength shall be determined as described in ASTM Standard D638. The tangent (approximating the initial slope) to the force-strain curve shall be obtained by drawing a straight line between the origin and the point on the curve corresponding to one-half the specified force (see table II). If the force strain-curve between these two points deviates by more than 0.1 percent elongation from the straight line drawn, a new straight line shall be drawn from the origin to one-quarter of the specified force. The 1.0 percent offset yield strength determined in this manner and the force corresponding to 3 percent elongation shall equal or exceed the value listed in table II for the tape being evaluated. In the event that the breakage of tape occurs prior to reaching the 1.0 percent offset yield strength or 3 percent elongation, the force at break shall equal or exceed the value listed in table II.

4.4.5 Elongation under stress.

4.4.5.1 Apparatus. The apparatus for this test shall consist of standard means of measuring length, positive nonslipping clamping devices, a 50 gram weight and a weight as specified in 4.4.5.3.

4.4.5.2 Preparation of samples. Samples shall be conditioned as specified in 4.4.1. Samples at least 24 inches long shall be clamped so as to hang in the test area (75° ± 5° F., 50 ± 5 percent r.h.) for at least 24 hours under no externally applied stress before tests are begun. Not less than five specimens of tape shall be tested.

4.4.5.3 Procedure. Before any weight is hung on the samples, a mark shall be made approximately 20 inches from the point of clamping to measure the elongation and recovery of the sample. The distance between the mark and clamping point shall be measured accurately to the nearest 0.01 of an inch with an applied tension of 50 grams. This distance shall be taken as the base distance for calculation of residual elongation. When the measurement of the base distance has been made, the test shall begin. A test weight of 20 pounds shall be attached to the tape below the mark at zero time and allowed to hang undisturbed for 180 minutes ± 30 seconds, at which time the weight shall be removed from the tape. The tape shall be allowed to hang under its own weight for an additional 180 minutes ± 30 seconds. The distance between the mark and the point of clamping shall then be measured to the nearest 0.01 inch with an applied tension of 50 grams. The difference between the base distance and the final distance shall be expressed as a percent of the base distance to determine compliance with 3.3.2.

4.4.6 Layer-to-layer adhesion.

4.4.6.1 Apparatus. The apparatus for this test shall consist of an 8,000 gram weight and a 1/2 inch diameter hollow tube. The tube shall be made of a nonoxidizing metal such as brass or corrosion resisting steel 1/2 inch in diameter and 4 inches in length, and shall weigh not less than 15 grams nor more than 30 grams. The tube shall be capable of being mounted in bearings so that it may be rotated freely around its central axis and easily removed from the bearings.

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4.4.6.2 Preparation of samples. No special preparation of the sample is required, except as specified in 4.4.1. Not less than five specimens of tape shall be tested.

4.4.6.3 Procedure. The test procedures shall be as follows:

(a) The 3 foot sample length of tape shall be fastened at one end, oxide side down, to the 1/2 inch diameter hollow tube with a nonoozing adhesive material. The tube shall then be mounted in the bearings such that the tape shall hang free below the tube.

(b) Attached to the free end of the tape shall be the 8000 gram weight. A small strip of double-coated adhesive tape shall be affixed to the oxide side of the tape one inch above the weight. The tube shall then be slowly and uniformly rotated so that the tape, held in tension by the weight, winds uniformly around the tube into a compact and even roll. The double-coated adhesive tape when wound into the test roll acts to secure the roll and prevent its unwinding when the weight is removed.

(c) The tube supporting the rolled tape shall be removed from the winding setup and subjected to a heat and humidity cycle in which the first 16 to 18 hours shall be at 130° F. and 85 ± 5 percent r.h., while the final 4 hours shall be at 130° F. dry heat (less than 5 percent r.h.). During the humidification and dry heat cycle, provision shall be made so that the air surrounding the tube is constantly circulated to assure uniformity of conditions throughout the test area. At the end of the dry heat cycle, the rod shall be removed from the conditioned area and allowed to come to equilibrium with room conditions, approximately 75° F. at 50 percent r.h.

(d) To evaluate the tape for layer-to-layer adhesion, the end of the roll on the rod shall be carefully opened and the double-coated tab shall be removed. The rod shall then be held between the thumb and fingers and the untabbed tape shall be observed to note if the first two or three layers loosen up of their own accord; if this occurs, there is obviously no adhesion and the tape has passed the test. If no loosening or very little loosening of the outermost layers is observed, the free end of the tape shall be unwound slowly until 9 inches has been unwound. The free end shall then be allowed to hang and the tape shall be observed to see if it will loosen by itself. If it will not unwind unaided, the rod, with the tape hanging freely, shall be slowly rotated in the direction of tape unwind. If the tape adheres to itself and refuses to begin to unwind after the rod has been rotated through one-fourth revolution or 90°, it shall be considered to have failed the test. After the rotation test has been made, the free end of the tape shall be held and the rod allowed to fall, thereby unwinding the tape. The unwound tape shall be checked for evidence of coating delamination and in this way the severity of adhesion is established. Any tape which will not self-unwind after rotating the rod through 90° or which shows any delamination, except in the 2 inches nearest the rod, shall be considered as having failed this test.

4.4.7 Longitudinal curvature.

4.4.7.1 Apparatus. No special apparatus, other than a 48 inch straightedge, is required.

4.4.7.2 Preparation of samples. No special preparation of the sample is required, except as specified in 4.4.1. Not less than five specimens of tape shall be tested.

4.4.7.3 Procedure. Curvature shall be determined by constraining a sufficient length of tape to lie in a horizontal plane under zero tension and by measuring the maximum deviation of either tape edge from a 48 inch straightedge.

4.4.8 Wound pile envelope. The tape shall be examined for the wound pile envelope defects listed in table VI to determine compliance with 3.3.5.

4.4.9 Anchorage. A visual examination shall be made of the tape after completion of all tests to determine compliance with 3.3.6.

4.4.10 Splices. The tape shall be examined to determine compliance with 3.3.7.

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4.4.11 Workmanship and general examination. The tape, as initially received from the manufacturer, shall be visually examined to determine that the reels and the tape are free from the applicable defects listed in table VI.

TABLE VI. General Examination Defects

Examine	Defect
Wind	Oxide surface not toward the hub.
Wound pile envelope	Tape pile not smoothly wound; loose, with visible folds, buckling, cinching, spoking, or gaps between the tape layers. Protruding tape edges, tape layers, or groups of layers. Plane of the tape pile not perpendicular to the reel axis. Tape not wound on reel that meets the requirements of 3.1.3.4.
Appearance	Tape surfaces not clean; presence of dirt, dust, lint, fuzz, or other foreign matter. Presence of blemishes, holes, tears, creases, or wrinkles, split or ragged edges.
Construction	Tape does not unroll evenly and uniformly; tape sticks; tape affected by commonly used solvents/ cleaners.
Condition, as received	Improper packaging. Improper marking. Damaged contents.

4.5 Magnetic tests and measurements.

4.5.1 Definitions.

4.5.1.1 Reference tape. An unrecorded length of tape (2,550 feet minimum) of the same type the magnetic tape manufacturer wishes to supply, shall be furnished to the General Services Administration for certification that the tape will meet all system performance specifications (including video and audio frequency response, video and audio signal-to-noise and audio distortion) of the reference test recorder. When found acceptable, this roll of tape will become the reference tape for that manufacturer's particular product type and it will be used for establishing the standard record conditions and the standard output levels for the reference test recorder when submitted samples of that particular tape type are tested as specified herein. The certification of output performance acceptability of the reference tape does not constitute product qualification as defined by 3.1.1.

4.5.1.2 Secondary reference tape. The secondary reference tape is an unrecorded length of tape, the magnetic characteristics of which have been calibrated against those of the reference tape. The secondary reference tape is used for preliminary adjustments of the reference test recorder and as a working reference.

4.5.1.3 Standard RF record conditions.

4.5.1.3.1 Standard RF output level. The standard RF output level is that reproduce level, before limiting and demodulation, of a 10 MHz signal which has been recorded on the reference tape as recommended by the manufacturer of the standard recorder.

4.5.1.4 Standard audio record conditions.

4.5.1.4.1 Operating bias current. Operating bias current is that bias current through the recording head which will give peak output from the reference tape at maximum audio record level (see 4.5.1.4.2).

4.5.1.4.2 Maximum audio record level. The maximum audio record level is that input level of a 1,000 Hz sinusoidal signal recorded on the reference tape at operating bias current such that on playback the output signal will have 3 percent third harmonic distortion.

4.5.1.4.3 Standard audio record level. The standard audio record level is 8 db below the maximum audio record level.

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4.5.1.4.4 Standard audio output level. The standard audio output level is that reproduce level of a 1,000 Hz signal which has been recorded on the reference tape in accordance with 4.5.1.4.3, standard audio record level.

4.5.1.5 Standard control track record conditions.

4.5.1.5.1 Standard control track record level. The standard control track record level is that input level of a 240 Hz signal for the reference tape, which is adjusted in accordance with SMPTE RP-16.

4.5.1.6 Standard cue track record conditions.

4.5.1.6.1 Operating bias current. Operating bias current is that bias current through the recording head which will give peak output from the reference tape at standard cue track record level (see 4.5.1.6.2).

4.5.1.6.2 Standard cue track record level. The standard cue track record level is that input level of a 1,000 Hz sinusoidal signal recorded on the reference tape at operating bias current such that on playback the output signal will have 3 percent third harmonic distortion.

4.5.2 Test equipment.

4.5.2.1 Reference test recorder. The reference test recorder shall be a 2 inch quadruplex, transverse-scan video tape recorder-reproducer which is capable of recording NTSC color/black and white video, audio and tracking control signals in accordance with ANSI Standards and SMPTE Recommended Practices (see 2.2). For qualification tests and the lot acceptance tests specified herein, the General Services Administration will use an Ampex VR-1200B video recording-reproducing system.

4.5.2.2 Test setup. The test setup for the reference test recorder prior to performing the magnetic tests shall be as follows:

(a) The recorder shall be thoroughly cleaned and demagnetized.
(b) The tensions of the recorder shall be measured to determine that the recorder is in proper adjustment.
(c) Locations and dimensions of the video, audio and tracking control records shall be in accordance with ANSI C98.6.
(d) Unless otherwise noted, all measurements shall be performed at a linear tape speed of 15 inches per second.
(e) For all tests, the reference test recorder shall be terminated with its proper load impedances.
(f) A take-up reel of the same diameter as the test tape reel shall be used.
(g) All audio magnetic requirements refer to Audio No. 1 Record (Program Audio) and all cue track requirements refer to Audio No. 2 Record (Cue Track).

4.5.3 Test procedures.

4.5.3.1 Sensitivity.

4.5.3.1.1 RF output sensitivity. The standard RF output level (see 4.5.1.3.1) shall be established to calibrate the reference test recorder for the sensitivity measurement. A 10 MHz signal shall then be recorded on the tape to be tested without any further recorder playback adjustments. This tape shall be reproduced and the output of the recorder measured to determine compliance with 3.4.1.1. The output shall be measured prior to any limiting or automatic gain control in the reproduce circuitry.

4.5.3.1.2 Audio sensitivity. The standard audio output level (see 4.5.1.4.4) shall be established to calibrate the reference test recorder for the sensitivity measurement. A 1,000 Hz sinusoidal signal shall then be recorded on the tape to be tested at the standard audio record level (see 4.5.1.4.3) and at operating bias current (see 4.5.1.4.1) and without any further recorder playback adjustments. This tape shall be reproduced and the output of the recorder measured to determine compliance with 3.4.1.2. The third harmonic distortion of the tape to be tested shall also be measured during this test to determine compliance to the requirement of table III.

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4.5.3.2 Frequency response.

4.5.3.2.1 Audio frequency response. The frequencies specified in table III shall be recorded on the reference tape at the standard audio record level (see 4.5.1.4.3) and at operating bias current (see 4.5.1.4.1). This tape shall then be played back and the output level at each frequency measured. Correction shall be made for the response of the reference test recorder with a secondary reference tape at operating bias current. This procedure shall be repeated on the tape to be tested. The response of the tested tape shall be compared to the response of the reference tape, after normalizing to 1,000 Hz to determine compliance with the requirements of table III.

4.5.3.3 Instantaneous nonuniformity (dropouts).

(a) For each particular class of tape, the dropout test shall be made on a virgin reel of tape, of a minimum length wound on a reel as specified in 3.1.3.4. For qualification purposes five sample reels of each class of tape submitted shall be subjected to the dropout test and at least four of the five samples shall meet the criteria specified in table III to be considered acceptable.

(b) A sync plus set-up (10 IRE) video signal shall be recorded over the entire length of tape. The video record head currents, tape tension, transport adjustments, control track record current and playback equalization shall be adjusted to those values which are optimum for the tape type being tested. The recorder shall be operated at 2.0 ± 0.1 mil tip penetration.

(c) Upon playback, all discontinuities in output, which for a period of 5 microseconds or more are equal to or exceed a 16 db decrease in average output shall be counted to determine compliance with the requirements of table III. The measurement shall be made prior to any limiting or automatic gain control in the reproduce circuitry. All dropouts occurring in the first and last 2 percent of the nominal tape length shall be disregarded.

4.5.3.4 Output level uniformity-short term.

4.5.3.4.1 Audio uniformity. The frequencies specified in table III shall be recorded at standard audio record level and with operating bias current over the entire tape length. The 1,000 Hz signal shall be recorded over the first half of the reel and the 8,000 Hz signal recorded over the second half of the reel. On playback, the output voltage variations shall be measured by means of a linear recording oscillograph having a response which is uniform to at least 200 Hz. The tape uniformity shall be measured by determining the maximum variation in output level. The tape shall meet the requirements specified in table III over any 10 second interval, with all non-recurring pips being disregarded. All output level variations occurring in the first and last 2 percent of the nominal tape length shall be disregarded.

4.5.3.4.2 Control track uniformity. A control track signal shall be recorded over the entire length of the tape at the standard control track record level (4.5.1.5.1). The frequency of the signal shall be 240 Hz. On playback, the output voltage variations shall be measured by means of a linear recording oscillograph which is uniform to at least 200 Hz. The tape uniformity shall be measured by determining the maximum variation in output level. The tape shall meet the requirements specified in table III over any 10 second interval, with all non-recurring pips being disregarded. All output level variations occurring in the first and last 2 percent of the nominal tape length shall be disregarded. This uniformity test may be performed simultaneously with the audio track uniformity tests (4.5.3.4.1 and 4.5.3.5.1) and instantaneous nonuniformity (dropouts) (4.5.3.3) tests specified herein.

4.5.3.4.3 Cue track uniformity. A 1,000 Hz sinusoidal signal shall be recorded at standard cue track record level (see 4.5.1.6.2) and with operating bias current (see 4.5.1.6.1) over the entire tape length. On playback, the output voltage variations shall be measured by means of a linear recording oscillograph having a response which is uniform to at least 200 Hz. The tape uniformity shall be measured by determining the maximum variation in output level. The tape shall meet the requirement specified in table III over any 10 second interval, with all non-recurring pips being disregarded. All output level variations occurring in the first and last 2 percent of the nominal tape length shall be disregarded. This uniformity test may be performed with the audio track uniformity, control track uniformity and instantaneous nonuniformity tests (dropouts) specified herein.

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4.5.3.5 Output level uniformity-long term.

4.5.3.5.1 Audio uniformity. The test procedure shall be similar to that specified in 4.5.3.4.1. The tape uniformity shall be measured by determining the variation of the peak output over the specified tape length. This uniformity test may be performed simultaneously with the output level uniformity-short term (4.5.3.4.1) and instantaneous nonuniformity (dropouts) (4.5.3.3) tests specified herein.

4.5.3.5.2 Control track uniformity. The test procedure shall be similar to that specified in 4.5.3.4.2. The tape uniformity shall be measured by determining the variation of the peak output over the entire length of tape.

4.5.3.5.3 Cue track uniformity. The test procedure shall be similar to that specified in 4.5.3.4.3. The tape uniformity shall be measured by determining the variation of the peak output over the entire length of tape.

4.5.3.6 Harmonic distortion.

4.5.3.6.1 Harmonic distortion at standard audio record level. The test procedure shall be as specified in 4.5.3.1.2.

4.5.3.6.2 Harmonic distortion at maximum audio record level. The test procedure shall be as specified in 4.5.3.1.2 except that the level of recording shall be maximum audio record level. The third harmonic distortion of the tape shall be measured to determine compliance to the requirement of table III.

4.5.3.7 Signal-to-noise ratio.

4.5.3.7.1 Video signal-to-noise ratio. A composite video gray level (50 IRE) signal shall be recorded on the reference tape. The video record head currents, vacuum guide position, tape tension, transport adjustments, control track record current and playback equalization shall be adjusted as recommended by the recorder manufacturer to those values which are optimum. The tape shall be played back and the RMS noise level, after removal of sync and blanking pulses, shall be measured. All noise components between 10 KHz and 4.2 MHz shall be measured. The above procedure shall be repeated on the tape to be tested and a comparison made to determine compliance with 3.4.7.

4.5.3.7.2 Audio signal-to-noise ratio. Using the reference tape, the reference test recorder shall be adjusted for operating bias current (see 4.5.1.4.1) and maximum audio record level (see 4.5.1.4.2). The reproduce equalization shall be adjusted as recommended by the recorder manufacturer. A 1,000 Hz sinusoidal signal shall be recorded and then reproduced with the playback gain adjusted to produce an output level of +8 dBm across the 600 ohm unbalanced output line. This reproduce output level shall be taken as the signal level. The tape shall then be externally erased. The input signal shall be disconnected and replaced with a load equal to the input impedance of the recorder, and the recorder operated in the record mode. The reproduced noise output level shall be measured without any readjustment of the recorder. The signal-to-noise ratio is the value in db of the signal level minus the value of the noise level. The signal-to-noise measurement shall be made utilizing a band pass filter which provides no more than a 3 db attenuation at the recorder band edge frequencies and has an 18 db per octave rolloff characteristic. The above measurements shall be repeated (without any recorder adjustment) on the tape being tested to determine conformance with 3.4.7.2.

4.5.3.8 Layer-to-layer signal transfer.

4.5.3.8.1 Audio signal transfer. The tape shall be externally erased. With the reference test recorder operating at maximum audio record level with operating bias current, and with erase head operating, ten layers of tape shall be recorded with no signal input to the reference test recorder and shall be wound on a reel classified as ER-2000-QCP in accordance with W-R-175. The record level shall be increased to maximum audio record level at operating bias current with a 1,000 Hz sinusoidal signal and one additional layer recorded and wound on the reel. The record level shall then be returned to zero and ten additional layers shall be recorded and wound on the reel. The recorded tape shall be conditioned at a temperature of 150 F. (dry heat) for a period of 4 hours.

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The tape shall then be played back and the output of the playback amplifier shall be connected to a 1,000 Hz band-pass filter having a 100 Hz passband. The level of the recorded signal and the maximum level of the signal resulting from signal transfer shall be measured at the output of the filter to determine compliance with 3.4.8.

4.5.3.9 Ease of erasure.

4.5.3.9.1 Audio erasure. The tape shall be externally erased. A frequency of 1,000 Hz shall be recorded at maximum audio record level and with operating bias current. The tape shall then be conditioned at a temperature of 150° F., (dry heat) for a period of 4 hours. The tape shall be reproduced, and the output measured through a properly terminated 1 kHz band pass filter having a 100 Hz passband. The tape shall then be erased on the reference test recorder with the bias current adjusted as recommended by the recorder manufacturer. The level of the residual signal on playback shall be measured, again through the band pass filter. The difference between the recorded signal level and the residual signal level is the effective reduction in signal or ease of erasure.

4.5.3.10 Electrical resistance.

4.5.3.10.1 Apparatus. The apparatus for this test shall consist of a temperature and humidity chamber and the apparatus required by Method 4041, Insulation Resistance for Flexible Tapes, of Fed. Test Method Std. No. 406.

4.5.3.10.2 Preparation of samples. Samples of virgin tape shall be prepared as specified under preliminary conditioning (see 4.4.1). Lengths of tape sufficient for this test shall be unwound from the reels and allowed to remain in a 50 ± 5 percent r.h. 75° ± 5° F. atmosphere without kinks or bends for at least 24 hours before test.

4.5.3.10.3 Procedure. The electrical resistance of either tape surface shall be determined in accordance with method 4041, Insulation Resistance for Flexible Tapes, of Fed. Test Method Std. No. 406 to determine conformance with 3.4.10. The samples shall be placed in a 30 percent r.h. or less, 70° ± 5° F. atmosphere for at least 6 hours prior to testing. The samples shall then be placed in the strip electrodes specified. The measurement potential shall be 500 ± 10 volts dc. Not less than five samples prior to the durability test (see 4.5.3.11) and not less than five samples which have been subjected to the durability test shall be measured. The average value of electrical resistance for each group shall be reported.

4.5.3.11 Durability. The test procedure shall be as follows:

(a) An initial recording of video, audio and control track signals in accordance with 4.5.3.3, 4.5.3.4.1 and 4.5.3.4.2 respectively shall be made over a two minute length of a virgin reel of tape wound on a reel as specified in 3.1.3.4. An initial dropout measurement of the reproduced RF signal shall be performed as specified in 4.5.3.3. An initial measurement of the output level uniformity-short term and long term, of the reproduced audio and control track signals shall be performed as specified in 4.5.3.4.1, 4.5.3.5.1, 4.5.3.4.2 and 4.5.3.5.2. The reproduced RF output level shall also be measured prior to any limiting in the reproduce circuitry.

(b) The tape shall then be subjected to 100 wear passes on the reference test recorder at the specified speed (see 4.5.2.2). A wear pass shall consist of a normal play pass and a high-speed rewind pass over the tape length being tested. After completion of the last wear pass, the tape path shall be cleaned and the uniformity, dropout, and RF output level measurements repeated to determine compliance with 3.4.11.

4.5.3.12 Abrasivity.

(a) The abrasivity test shall be made on a virgin reel of tape, of a minimum length wound on a reel as specified in 3.1.3.4.

(b) A sync plus set-up (10 IRE) video signal shall be recorded over the entire length of tape. The video record current, tape tension transport adjustments, control track record current and playback equalization shall be adjusted to those values which are optimum for the tape type being tested. The recorder shall be operated at a 2.0 ± 0.1 mil tip penetration with a head having a tip projection of between 2.4 and 3.0 mils.

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(c) This test may be performed in conjunction with the instantaneous nonuniformity (dropouts) test. The recorded reel of tape shall be subjected to a total of 20 hours of play mode operation on the reference test recorder at the specified speed (see 4.5.2.2). Testing shall consist of a normal play pass and a high speed rewind pass over the entire reel of tape. The tip projection above the drum shall be measured prior to the initial pass and subsequent to the 20th hour. The head wear rate shall be calculated as the difference between tip projection measurements divided by the total play time of the reel of tape. The average wear rate of the four head tips shall meet the requirements of 3.4.12.

4.5.3.13 Environmental extremes. An initial recording of video, audio and control track signals in accordance with 4.5.3.3, 4.5.3.4.1 and 4.5.3.4.2 respectively shall be made over the entire tape length of a virgin reel of tape wound on a reel as specified in 3.1.3.4. The tape shall then be subjected to a temperature of 120° ± 5° F. and a relative humidity of 80 to 90 percent for a period of 8 hours. Upon removal from this environment, the tape shall be rewound once on the reference test recorder and kept at a temperature of 75° ± 5° F. and a relative humidity of 50 ± 5 percent for a minimum of 16 hours. The tape shall then be tested for instantaneous nonuniformity (dropouts) (see 4.5.3.3), output level uniformity-short term (see 4.5.3.4.1 and 4.5.3.4.2) and output level uniformity-long term (see 4.5.3.5.1 and 4.5.3.5.2) to determine compliance with 3.4.13. Following these tests, the tape shall then be subjected to a temperature of 10° ± 5° F. and a relative humidity of less than 10 percent for a period of 8 hours. Upon removal from this environment, the tape shall be rewound once on the reference test recorder and kept at a temperature of 75° ± 5° F. and a relative humidity of 50 ± 5 percent for a minimum of 16 hours. The tape shall again be tested for instantaneous nonuniformity (dropouts) (see 4.5.3.3), output level uniformity-short term (see 4.5.3.4.1 and 4.5.3.4.2) and output level uniformity-long term (see 4.5.3.5.1 and 4.5.3.5.2) to determine compliance with 3.4.13.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.1).

5.1.1 Level A. The reels of tape shall be individually enclosed in close-fitting heat sealable plastic bags, 3 to 4 mils thick, the material for these bags shall conform to L-P-378, type I, grade B, finish 1. Included air volume of the sealed bags shall be kept to a practicable minimum.

5.1.2 Level C. The reels of tape shall be individually packaged in accordance with the manufacturer's standard practice providing that it will insure protection for the tape during shipment andsafe delivery to its destination.

5.2 Intermediate packaging. Intermediate packaging shall be level A, B, or C, as specified (see 6.1).

5.2.1 Level A. Each packaged reel of tape shall be further packaged in a close-fitting box conforming to PPP-B-636, class weather-resistant, style optional. Each box shall be provided with a centering hub constructed from a material adequate to provide support for the reel by the hub in such a manner as to suspend the reel in the box to prevent "resting" on the flanges or movement within the closed box. The boxes shall be closed in accordance with the appendix to PPP-B-636.

5.2.2 Level B. The packaged reels shall be further packaged in accordance with 5.2.1 except that the fiberboard boxes shall be class-domestic.

5.2.3 Level C. The reels of tape packaged as specified in 5.1.2 shall be further packaged in accordance with the manufacturer's standard practice providing that it will insure protection for the tape during shipment and safe delivery to its destination.

5.3 Packing. Packing shall be level A, B, or C, as specified (see 6.1).

5.3.1 Level A. The tape packaged as specified herein shall be packed in multiples in a close-fitting box conforming to PPP-B-636, class weather-resistant. The shipping container shall be closed, sealed, and strapped in accordance with the appendix to PPP-B-636.

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5.3.2 Level B. The tape packaged as specified herein shall be packed in quantities as specified in the contract or order in accordance with 5.3.1, except that the fiberboard boxes shall be class-domestic.

5.3.3 Level C. Reels of tape packaged as specified in 5.2.3 shall be packed in containers that will assure carrier acceptance and safe arrival at destination in compliance with the Uniform Freight Classification Rules or the National Motor Freight Classification Rules.

5.4 Marking. The marking shall be in accordance with 5.4.1 or 5.4.2, as specified (see 6.1).

5.4.1 Civil agencies. In addition to the special markings required by 5.4.3 and any special marking required by the contract or order, all intermediate and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.4.2 Military agencies. In addition to the special markings required by 5.4.3 and any special markings required by the contract or order, all intermediate and shipping containers shall be marked in accordance with MIL-STD-129.

5.4.3 Special marking.

5.4.3.1 Intermediate container. In addition to the marking required by 5.4.1 or 5.4.2, the intermediate package shall be permanently marked with the following information:

(a) Tape, Magnetic Oxide Coated, Broadcast Video.
(b) Classification.
(c) Name of manufacturer.
(d) Manufacturer's identifying tape name/number.
(e) Production date (5 digit number with first 2 digits indicating year of production and last 3 digits indicating day of production completion).
(f) Production code or serial number (unique code by which the manufacturer identifies the product batch, coater, slitter and web position of the tape).
(g) Federal Stock No. (if available).
(h) Lot number.

5.4.3.2 Shipping containers. In addition to the marking required by 5.4.1 or 5.4.2, each shipping container shall be marked with the following information:

(a) Tape, Magnetic Oxide Coated, Broadcast Video.
(b) Classification.
(c) Manufacturer's identifying tape name/number.
(d) Production date.
(e) Lot number.
(f) Contract number.
(g) Federal Stock number (if available).

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Quantity of tape required.
(c) Tape designator (see 1.2.1, 1.2.2, and 1.2.3).
(d) Width required (see 1.2.4 and 3.2.2).
(e) Reel required (see 1.2.5, 1.2.6, and 3.1.3.4).
(f) Length required (see 1.2.7 and 3.2.1).
(g) Inspection responsibility (see 4.1).
(h) Packaging and packing required (see 5.1, 5.2 and 5.3).
(i) Whether civil or military marking is required (see 5.4.1 and 5.4.2).

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6.2 With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in Qualified Products List QPL-W-T-001567, whether or not such products have actually been so listed by that date. The attention of suppliers is called to this requirement and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Federal Supply Service, General Services Administration, Washington, DC 20406, and information pertaining to qualification of products may be obtained from that activity. Application for qualification tests shall be made in accordance with "Provisions Governing Application by Manufacturers for Inclusion on Federal Qualified Products Lists (QPL's)" (see 6.3).

6.3 Copies of "Provisions Governing Application by Manufacturers for Inclusion on Federal Qualified Products Lists (QPL's)" may be obtained upon application to the Assistant Commissioner, Office of Standards and Quality Control (FMH), Federal Supply Service, General Services Administration, Washington, DC 20406.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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